In other pages, we obtained one of the greatest results,
the primary decomposition.
Now, we will walk into deeper result, the second decomposition theorem.
in the right hand is the primary decomposition,
in the left hand is the observation for cyclic subspace.

「The primary decomposition theorem.
Let T be a linear operator on the finite-dimensional vector space V over a field F.
Let P be the minimal polynomial for T with the irreducible decomposition

$$P = P_1^{r_1} \cdot P_2^{r_2} \cdots P_k^{r_k}$$

Then,

$V = \bigoplus_{i=1}^{k} \ker P_i^{r_i}(T)$ and each $\ker P_i^{r_i}(T)$ is invariant T.

Moreover, the restriction of T by $\ker P_i^{r_i}(T)$ has the minimal polynomial as $P_i^{r_i}$.

「Thm. Let α be any non-zero vector in V and let P_α be the T-annihilator of α.

1) $\deg P_\alpha = \dim Z(\alpha, T)$
2) $Z(\alpha, T) = \mathrm{span}\{T^k\alpha \mid 0 \le k \le \deg P_\alpha\}$
3) The minimal polynomial of $T|_{Z(\alpha,T)}$ is P_α.

「Cyclic decomposition Theorem.

「Thm. Let $T: V \to V$ be a linear operator on finite-dimensional space V. Suppose that the minimal polynomial of T has a irreducible decomposition

$$m_T(x) = p_1(x)^{r_1} \cdots p_k^{r_k}$$

Then, by the primary decomposition, V is decomposed by T-invariant spaces

$$V = \ker p_1^{r_1} \oplus \cdots \oplus \ker p_k^{r_k}$$

With $T|_{\ker p_i^{r_i}}$ has minimal polynomial as $p_i^{r_i}$」 여기까지가 제1 분해.

Now, for fixed $1 \le j \le k$, denote $T_j = T|_{\ker p_j^{r_j}}$ and $W_j = \ker p_j^{r_j}$.
Then, there are non-zero vectors $\alpha_1, \cdots, \alpha_n$ such that

$$W_j = Z(\alpha_1, T_j) \oplus \cdots \oplus Z(\alpha_n, T_j)$$

satisfying for some unique integers $1 \le r_n \le \cdots \le r_2 \le r_1 = r_j$,

$$m_{T_j|Z(\alpha_s, T_j)} = p_j^{r_s}.$$

「Computation」

Let A be a 8 by 8 Matrix over $\mathbb{Q}$ as

$$A = \begin{bmatrix} 1 & 1 & & & & & & \\ 0 & 1 & & & & & & \\ & & 1 & 0 & 0 & & & \\ & & 0 & 0 & -1 & & & \\ & & 0 & 1 & 0 & & & \\ & & & & & 0 & 0 & 2 \\ & & & & & 1 & 0 & 0 \\ & & & & & 0 & 1 & 0 \end{bmatrix}$$

Denote $A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$, $A_3 = \begin{bmatrix} 0 & 0 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

Find the Rational form.

1. The characteristic Polynomial is

$$[(t-1)^2-0]\cdot[(t-1)\cdot(t^2+1)]\cdot[t^3-2] = (t-1)^3\cdot(t^2+1)\cdot(t^3-2).$$

2. The minimal polynomial is: Since

$$A^2 = \begin{bmatrix} A_1 & & \\ & A_2 & \\ & & A_3 \end{bmatrix}^2 = \begin{bmatrix} A_1^2 & & \\ & A_2^2 & \\ & & A_3^2 \end{bmatrix},$$

So the minimal polynomial of A is the least common multiple,

$$m_A = \operatorname{lcm}(m_{A_1}, m_{A_2}, m_{A_3}) = (t-1)^2\cdot(t^2+1)\cdot(t^3-2).$$

By the primary decomposition theorem,

$$V = \operatorname{Ker}(A-I)^2 \oplus \operatorname{Ker}(A^2+I) \oplus \operatorname{Ker}(A^3-2I)$$

Meanwhile, $\dim \operatorname{Ker}(A-I)^2 = \deg \chi_{A_1} = 3$, and the calculation gives

$\operatorname{Ker}(A-I)^2 = \operatorname{Span}\{e_1, e_2, e_3\}$ and $Ae_2 = e_1 + e_2$, so $Z(e_2, A) = \operatorname{Span}\{e_2, Ae_2\}$
$= \operatorname{Span}\{e_1, e_2\}.$

and, $Ae_3 = e_3$, so $Z(e_3, A) = \operatorname{Span}\{e_3\}$, the annihilator $\operatorname{Ann}(e_3, A) = t-1$.

Therefore, $\operatorname{Ker}(A-I)^2 = Z(e_2, A) \oplus Z(e_3, A)$

with the minimal polinomial $(t-1)^2$, $t-1$.

「Debris」

$$A^2 + I = \begin{bmatrix} 1 & 2 & & & & & & \\ 0 & 1 & & & & & & \\ & & 1 & 0 & 0 & & & \\ & & 0 & -1 & 0 & & & \\ & & 0 & 0 & -1 & & & \\ & & & & & 0 & 2 & 0 \\ & & & & & 0 & 0 & 2 \\ & & & & & 1 & 0 & 0 \end{bmatrix}$$

first, observe that

$$= \begin{bmatrix} 2 & 2 & & & & & & \\ 0 & 2 & & & & & & \\ & & 0 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 1 & 2 & 0 \\ & & & & & 0 & 1 & 2 \\ & & & & & 1 & 0 & 1 \end{bmatrix} + I_8$$

$$A^3 - 2I = \begin{bmatrix} 1 & 3 & & & & & & \\ 0 & 1 & & & & & & \\ & & 1 & 0 & 0 & & & \\ & & 0 & 0 & 1 & & & \\ & & 0 & -1 & 0 & & & \\ & & & & & 2 & 0 & 0 \\ & & & & & 0 & 2 & 0 \\ & & & & & 0 & 0 & 2 \end{bmatrix} - 2\cdot I_8 = \begin{bmatrix} -1 & 3 & & & & & & \\ 0 & -1 & & & & & & \\ & & -1 & 0 & 0 & & & \\ & & 0 & -2 & 1 & & & \\ & & 0 & -1 & -2 & & & \\ & & & & & 0 & & \\ & & & & & & 0 & \\ & & & & & & & 0 \end{bmatrix}$$

Similarly, $\ker(A^2+I) = \operatorname{span}\{e_4, e_5\} = Z(e_4, A)$, since $Ae_4 = e_5$.

$\ker(A^3-2I) = \operatorname{span}\{e_6, e_7, e_8\} = Z(e_6, A)$, since $Ae_6 = e_7$, $A^2e_6 = e_8$.

Consequently,

$$V = Z(e_3, A) \oplus Z(e_2, A) \oplus Z(e_4, A) \oplus Z(e_6, A)$$

with $(t-1)$ $\quad (t-1)^2 \quad t^2+1 \quad t^3-2$.

Now, the rational form is:

$$[A]_{\mathcal{C}} = \begin{bmatrix} 1 & & & & & & & \\ & 0 & & & & & & 2 \\ & 1 & 0 & & & & & -4 \\ & & 1 & 0 & & & & 4 \\ & & & 1 & 0 & & & -5 \\ & & & & 1 & 0 & & 4 \\ & & & & & 1 & 0 & -2 \\ & & & & & & 1 & 2 \end{bmatrix}$$

「Theoritical Exercises in HK.

「7.2.2. Let T be a linear operator on the fin. dim V. Then,

$\ker T$ and $\operatorname{Im} T$ are independent if and only if

$\ker T$ is the unique T-invariant Complementary of $\operatorname{Im} T$.

Proof. Suppose that $\ker T$ and $\operatorname{Im} T$ are independent.
Given $\alpha \in \ker T \cap \operatorname{Im} T$, $\alpha + (-\alpha) = 0$ and $-\alpha \in \ker T \cap \operatorname{Im} T$,
thus the independentness gives $\alpha = 0$. Therefore, $\ker T \cap \operatorname{Im} T = \{0\}$. 」General Property of independence.
Let $\{\alpha_1, \alpha_2, \cdots, \alpha_k\}$ be a basis for $\operatorname{Im} T$.
And, by the dimension theorem, the number of elements of $\ker T$ is $n-k$,
So let $\{\alpha_{k+1}, \cdots, \alpha_n\}$ be the basis for $\ker T$. 」Dimension theorem 은 쓰이라
Again, the independentness gives that $\{\alpha_1, \cdots, \alpha_k, \alpha_{k+1}, \cdots, \alpha_n\}$ are
linearly independent, so this is basis for V, which Proves that

$$V = \ker T \oplus \operatorname{Im} T.$$

The T-invariantness is obtained by: $T(\alpha) = 0 \Rightarrow T(T(\alpha)) = T(0) = 0$. 」
To show the uniqueness, let $W < V$ satisfies $V = W \oplus \operatorname{Im} T$ and $T[W] \subseteq W$.
Given $\alpha \in W$, $T(\alpha) \in W \cap \operatorname{Im} T = \{0\}$, so $\alpha \in \ker T$. 」이건 independence랑 상관 X.
Therefore, $W \subseteq \ker T$ and since $\dim W = \dim V - \dim \operatorname{Im} T = \dim \ker T$,
So $W = \ker T$.

The Converse is fucking clear.

Thm. 2.5. Let T be a linear operator on the vector space V over F. Suppose $V_1, \cdots, V_k < V$ are T-invariant subspaces and $V = V_1 \oplus V_2 \oplus \cdots \oplus V_k$. Given a polynomial $f \in F[t]$,

$$f(T)[V] = f(T)[V_1] \oplus \cdots \oplus f(T)[V_k].$$

Moreover, if the T-annihilator of α and β are same, then the T-annihilator of $f(T)(\alpha)$ and $f(T)(\beta)$ are same.

Proof. Given $\alpha \in V$, α has a unique representation

$$\alpha = \alpha_1 + \cdots + \alpha_k \text{ where } \alpha_i \in V_i.$$

Then, $$f(T)(\alpha) = f(T)(\alpha_1) + \cdots + f(T)(\alpha_k)$$

and each $f(T)(\alpha_i) \in W_i$, since each V_i is T-invariant.
And, suppose that for some $f(T)(\beta_i) \in f(T)[V_i]$ $(1 \le i \le k)$,

$$f(T)(\alpha_1) + \cdots + f(T)(\alpha_k) = f(T)(\beta_1) + \cdots + f(T)(\beta_k).$$

Then

$$f(T)(\alpha_1 - \beta_1) + \cdots + f(T)(\alpha_k - \beta_k) = 0$$

and $\alpha_i - \beta_i \in V_i$, so $f(T)(\alpha_i - \beta_i) \in V_i$, so

$$f(T)(\alpha_1 - \beta_1) = \cdots = f(T)(\alpha_k - \beta_k) = 0,$$

It proves the representation of $f(T)(\alpha)$ is unique. Consequently

$$f(T)[V] = f(T)[V_1] \oplus \cdots \oplus f(T)[V_k].$$ It proves the first.

Now, let $p(t) \in F[t]$ be the T-anihilator of α and β.
That is, $g(T)(\alpha) = 0$ and $h(T)(\alpha) = 0$ implies $p(t) \mid g(t), h(t)$.
Now, suppose that $r(t) \in F[t]$ satisfies that $r(T)(f(T)(\alpha)) = 0$.
Then, $r(T)f(T)(\alpha) = f(T)r(T)(\alpha) = 0$, so that $p(t) \mid f(t) \cdot r(t)$.

7.2.11.

Prop. Let A and B be 3 by 3 matrices over a field F. Then, A is similar to B if and only if $\chi_A = \chi_B$ and $m_A = m_B$ where χ and m are the Charateristic and the minimal, respectively.

Prof. For a Cubic Polynomial f, there are only three possible cases:

f is irreducible or f has a linear factor and degree 2 irreducible factor

or f splits over F into three linear factors.

Assume $\chi_A = \chi_B$ and $m_A = m_B$. Simply, denote $\chi = \chi_A$ and $m = m_B$.

Suppose that χ is irreducible. Then, since the minimal polynomial divides the characteristic, so $\chi = m$. Now, A and B both are similar to the matrix

$$C = \begin{bmatrix} 0 & 0 & -c_0 \\ 1 & 0 & -c_1 \\ 0 & 1 & -c_2 \end{bmatrix}$$ where $\chi(t) = t^3 + c_2t^2 + c_1t + c_0$.

Suppose that χ factors into a linear factor and a degree 2 irreducible factor. But, since the minimal polynomial divides the characterstic, m is either the linear or the degree 2 factor or f itself.

But, the minimal polynomial is linear implies that T is just multiple of identity, so the characteristic cannot have irreducible of degree 2, so it cannot happen. And, if the minimal polynomial is degree 2 irreducible, but since m and χ shares the roots, so m has a root in the given field, so it cannot be irreducible degree 2. Therefore, $m = \chi$ which implies that $A \sim C$ and $B \sim C$ where C is a Matrix above, So $A \sim B$.

Suppose that χ factors into three linear factors, as $\chi = (t-a)(t-b)(t-c)$.

If the minimal has degree 1, WLOG, $m = t-a$, then $A = B = aI$.

If the minimal has degree 2, WLOG, $m = (t-a)(t-b)$, then $c = a$ or b, and then $A, B \sim \begin{bmatrix} c & & \\ & 0 & -ab \\ & 1 & a+b \end{bmatrix}$

Finally, if $m = \chi$, then again $A, B \sim C$. It completes the Classifi.